

Decision-Focused Learning for Kidney Exchange Programs

When Does It Help?

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Final Internship Presentation

Core question: can we explain when decision-focused learning helps, fails, or becomes unnecessary in kidney exchange?

From “Can DFL Help?” to “When Does DFL Help?”

First presentation

Question

Can decision-focused learning improve controlled KEP decisions?

Finding

Promising signal, but not robust enough yet.

Final presentation

Question

When does DFL help, fail, or become unnecessary?

Answer

It is a candidate-reranking effect inside the feasible exchange landscape.

Main shift

From **method comparison** to **mechanism understanding**.

Recap: Kidney Exchange as a Weighted Decision Problem

Input

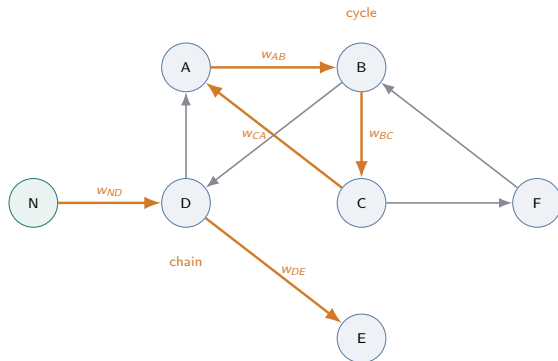
A directed compatibility graph with recipient-donor pairs and non-directed donors.

Decision

Select vertex-disjoint cycles and chains under length limits.

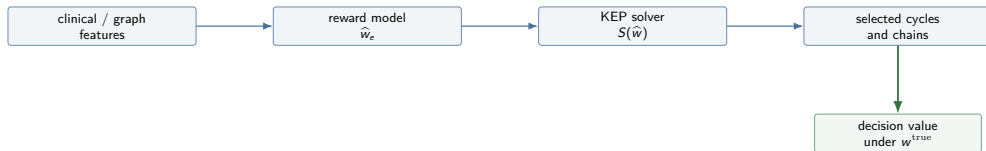
Objective

Maximize total predicted transplant benefit.



The learning model predicts edge weights; the optimizer chooses the exchange plan.

Prediction Accuracy Is Not the Final Goal



Two-stage learning

Fit edge rewards first, then optimize.

Decision-focused learning

Train for downstream decision quality.

New observation

In KEP, a **different** selected plan is not always a **bad** selected plan.

Key Hypothesis: Close Substitute Solutions

Hypothesis

Kidney exchange often contains many **close substitute exchange plans**.

Why KEP can behave this way

- A feasible plan is a packing of cycles and chains.
- One exchange can often be replaced by another.
- The plan identity changes, but its objective value may stay close.

Oracle plan

Cycle A + Chain B + Cycle C

} small replacement

Two-stage plan

Cycle A + Chain B' + Cycle C

Simple implication

$$\text{gap}(y_{2\text{stage}}) \leq \epsilon$$

$\implies$ max possible improvement $\leq \epsilon$.

Interpretation

DFL can only remove regret that is actually present.

Looking Beyond the Best Predicted Solution

Diagnostic idea

For each trained model and graph:

- 1 Find the best predicted KEP solution.
- 2 Exclude it with a no-good cut.
- 3 Re-solve for the next distinct solution.
- 4 Evaluate both under the true synthetic reward.

Question

If rank 1 is removed, does the next solution collapse, or remain near-oracle?

Predicted candidate ranking

rank 1 selected by model

rank 2 next distinct alternative

rank 3 another candidate

rank 20 deeper candidate basin

→ true value

Finding 1: Many Second-Best Solutions Are Still Good

2stage rank 2

7.04%

mean normalized oracle gap

45.19%

within 5% of oracle

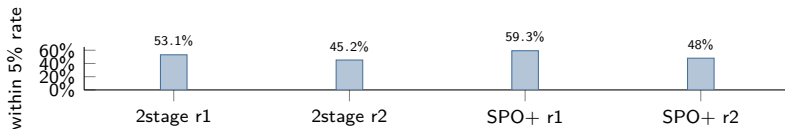
SPO+ rank 2

6.32%

mean normalized oracle gap

47.97%

within 5% of oracle



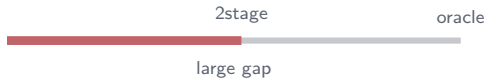
Interpretation

Excluding the best predicted solution often still leaves a near-oracle alternative.

Why Average DFL Gains Can Be Moderate

When two-stage is bad

DFL has room to help.



When two-stage is close

Little regret remains to remove.



Ceiling effect

If the two-stage plan is already a close substitute, even a perfect DFL method can only improve a little.

This explains why KEP can have many solution-identity errors but only moderate average regret improvements.

Finding 2: SPO+ Often Reranks Existing Candidates

Two-stage predicted ranking

rank 1 A: selected by two-stage

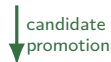
rank 2 ...

rank 8 B: near-oracle candidate

rank 20 ...

SPO+ selected ranking

rank 1 B: promoted by SPO+



Successful reranking

The promoted candidate is near-oracle.

Harmful reranking

The promoted candidate is not near-oracle.

Same mechanism; different candidate quality.

Case Examples: Correction, Promotion, and Harmful Reranking

G-1285

Clean rank-2 promotion

- SPO+ rank 1 equals two-stage rank 2.
- Promoted solution equals true oracle top 1.
- Cleanest success case.

G-392

Deep-candidate correction

- Two-stage contains a near-oracle candidate.
- But it appears around rank 8.
- SPO+ promotes it to rank 1.

G-14

Harmful reranking

- SPO+ also promotes a lower-ranked candidate.
- But the promoted candidate is worse under true reward.
- Reranking is risky.

Mechanism: SPO+ does not simply “find new topology.” It changes the ranking of feasible exchange candidates.

Ambiguity Means Opportunity and Risk

Prediction-boundary ambiguity

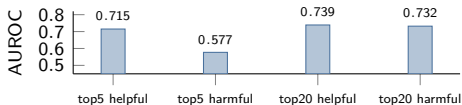
- small top-1 / top- k margin
- many near-tied candidates
- diverse top- k solutions
- unstable rank-1 identity across seeds

Interpretation

High ambiguity means DFL reranking can matter.

But

It does not say whether reranking will help or hurt.



Takeaways: DFL Is Useful, but Conditional

What worked

- DFL can improve controlled KEP decisions.
- SPO+ gives stronger held-out gains than the earlier FY probe.
- It can correct decision-critical ranking errors.

What changed

- The main question is no longer only: "Does DFL beat MSE?"
- It is: "When is there regret left for DFL to remove?"
- Mechanism matters as much as average performance.

What remains

- clinical reward models
- prospective diagnostics
- cross-regime robustness
- safer use when reranking is risky

Conclusion: In KEP, DFL is best understood as mechanism-aware candidate reranking, not as a universal replacement for predict-then-optimize.

Backup: KEP Mathematical Setup

Compatibility graph

$$G = (\mathcal{V}, \mathcal{A}), \quad \mathcal{V} = \mathcal{R} \cup \mathcal{N}$$

- $\mathcal{R}$: incompatible recipient–donor pairs
- $\mathcal{N}$: non-directed donors
- $(u, v) \in \mathcal{A}$: donor at u can donate to recipient at v

Decision oracle

$$S(w) \in \arg \max_{y \in \mathcal{Y}(G; K, L)} w^\top y$$

- K : maximum cycle length
- L : maximum chain length
- y : selected edge-incidence vector

Learning challenge

The oracle is combinatorial and piecewise constant in the predicted weights.

Backup: Synthetic-Label Decision Gap

Evaluation metric

$$\begin{aligned}y^{\star} &= S(w^{\text{true}}), & \hat{y} &= S(\hat{w}) \\ \text{gap}(\hat{y}) &= (w^{\text{true}})^{\top} y^{\star} - (w^{\text{true}})^{\top} \hat{y} \\ \text{normalized gap} &= \frac{(w^{\text{true}})^{\top} y^{\star} - (w^{\text{true}})^{\top} \hat{y}}{|(w^{\text{true}})^{\top} y^{\star}|}\end{aligned}$$

Important boundary

The “true” reward here is a synthetic benchmark label, not a clinical outcome.

Backup: Robustness Lessons from the First Generalization Study

Unseen 1000 graphs

- Strong FY gains on the original held-out split did not replicate cleanly.
- Two-stage baseline was very stable across train sizes.
- Many methods selected identical KEP decisions on most unseen graphs.

Realistic-synthetic label shift

- Two-stage was tied or slightly better across completed train sizes.
- Controlled noisy-linear training did not transfer cleanly to the more complex synthetic reward definition.

Lesson: the project needed mechanism analysis, not only larger performance tables.

Backup: Strict SPO+ Surrogate Ablation

Golden rule

Only change the surrogate; keep the data, split, model class, initialization, training budget, checkpoint protocol, and evaluation metrics fixed.

train size	2stage gap	SPO+ gap	improvement	95% CI
50	0.8480	0.7401	0.1079	[0.0347, 0.1960]
200	0.8451	0.7706	0.0745	[0.0170, 0.1527]
600	0.8223	0.7706	0.0517	[0.0044, 0.1112]
1200	0.8441	0.7846	0.0594	[0.0033, 0.1312]

Reading the table

Held-out400 results for the validation-SPO+-loss checkpoint; lower gap is better.

Shortest-path benchmark

- Oracle/objective alignment
- two-stage LR alignment
- SPO+ forward-loss alignment
- gradient and one-step SGD update alignment

Differences: around 10^{-6} or below

KEP integration preflight

- reward-max / cost-min adapter
- same cached KEP solver wrapper
- loss and gradient agreement
- graph-specific edge-selection vectors

Phase-0 max absolute diffs: 0

Backup: Cycle-Length Sensitivity

K	method	rank-2 mean gap	rank-2 median gap	within 5%
3	2stage	7.04%	5.75%	45.19%
4	2stage	6.93%	5.53%	47.50%
5	2stage	6.88%	5.20%	48.94%
3	SPO+	6.32%	5.28%	47.97%
4	SPO+	6.20%	5.03%	49.78%
5	SPO+	5.61%	4.39%	54.83%

Interpretation

The close-rank-2 behavior is not only an artifact of the max-cycle-3 feasible set.

Backup: Mechanism Atlas for Selected Cases

graph	updated mechanism	B found	SPO+=B	C rank	true Δ A→C
G-392	deep-candidate correction / reranking	1.00	1.00	8	+61.58
G-1285	clean exact rank-2 promotion	1.00	1.00	2	+42.02
G-1560	large-effect top-K promotion	1.00	1.00	2	+70.42
G-1169	broad top20 promotion	1.00	0.20	11	+29.17
G-1449	deep top20 promotion outside top5	1.00	0.00	8	+20.84
G-14	harmful SPO+ reranking	1.00	0.00	9	-28.14
G-163	harmful SPO+ reranking	1.00	0.00	8	-44.57

Notation

A = two-stage rank 1; B = best near-oracle candidate in two-stage top20; C = SPO+ rank 1.

Backup: Evidence Boundaries

Prospective graph features

- density
- degree distribution
- component structure
- cycle counts

Weak single-feature rules

Feasible-set geometry

- exchange candidates
- chain/cycle structure
- conflict graph
- participation features

Mechanism-relevant

Prediction-boundary features

- top- k margins
- near-tied counts
- top- k diversity
- ambiguity score

Post-training diagnostics

Avoid claiming that raw topology causes SPO+ success; the safer claim is feasible-landscape-conditioned DFL gain.

Scientific next steps

- richer clinical edge-benefit models
- graft survival / QALY / fairness integration
- cross-regime robustness beyond one Step2c setting
- prospective, pre-training suitability diagnostics

Method next steps

- stronger graph predictors
- better validation protocol for reranking risk
- multivariate diagnostic models
- efficient solver-in-the-loop training

Clinical boundary

Current results are synthetic decision-learning diagnostics, not yet clinical deployment evidence.